Cary Li

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EDUCATION

University of Toronto - St. George Campus Sep. 2026 - Present
B.A.Sc. Engineering Science | Toronto, ON

Dr. G.W. Williams Secondary School Sep. 2022 - May 2026
IB Diploma: 41/45 | Ontario Secondary School Diploma | Aurora, ON

PROJECTS

Kera | Hack the Valley 11 Oct. 2026
Python, PyTorch, scikit-learn

- Trained a three-seed PyTorch MLP ensemble on 42 colour features to estimate hemoglobin from fingernail photos; achieved 15.96 g/L mean absolute error on 50 held-out patient records.
- Built the colour-feature preprocessing and inference pipeline for the live app in 36 hours; trained with AdamW, Huber loss and five-fold grouped cross-validation.

ClearPath
TypeScript, React, Next.js, Three.js

- Built a browser-based 3D room planner with synchronized 3D and top-down views, drag-and-drop furniture and a first-person viewer to compare layouts before moving anything.
- Added a sunlight simulator with a heat map to find the best napping spots, and generated 3D models of IKEA and Walmart furniture to place in the room.

Anchor | CS50 Final Project Jun. 2026 - Present
Python, Flask, SQLite, Electron, JavaScript, Tailwind CSS

- Developed a desktop habit tracker with daily check-ins and streaks, built around three core workflows: habits, reflections and a goal timeline.
- Built the Flask and SQLite backend, packaged the app with Electron and released a web demo.

Racing Algorithms | UofT Formula Racing and Blue Sky Solar Racing Sep. 2026 - Present
C++, ROS 2, Foxglove | LiDAR perception and race strategy

- Project lead in UTFR cone detection: building a lightweight ML model that predicts cone position from LiDAR points.
- Developing an energy simulator for Blue Sky's Gen 13 solar car to search for efficient racing speeds.

EXPERIENCE

Co-Founder | The Fundamental Why Mar. 2025 - Present

- Created a physics and philosophy podcast with 7 episodes, 2,100+ subscribers and 14,100+ views.
- Interviewed STEM researchers, physics students and a nonprofit founder across YouTube and podcast platforms.

Competitive Swim Instructor | Champion Swim School Sep. 2024 - Jun. 2026

- Taught all four competitive strokes to ages 6-16 in classes of six; designed drills for 9.5 teaching hours per week.
- Named Instructor of the Month among 35 instructors in October 2025 and February 2026.

AWARDS

UofT National Book Award (2026): Sole recipient among 387 graduating students, recognized for academic achievement, creative thinking and breadth of extracurricular involvement.
Pay It Forward Scholarship (2026): Awarded $1,000 as one of the top five community leaders in Aurora and Newmarket by Magna International Inc. and Neighbour Media.
DECA ICDC Qualifier (2024): Led a team of three on Terra Fuel, a biomass-to-energy concept using pyrolysis and carbon filtration to target net-zero emissions; advanced past 300+ competitors to the international stage.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript, SQL, Java, C, C#, HTML/CSS
Frameworks and tools: PyTorch, scikit-learn, ROS 2, Foxglove, React, Next.js, Three.js, Flask, Electron, SQLite, Git